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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR/IMPROVEMENT/REAPPEARANCE EXAMINATION APRIL 2026

SECOND SEMESTER

Multi-Disciplinary Courses (MDC) - MG2MDCBCH100 - BIOCHEMISTRY IN ENTREPRENEURSHIP
(2024 ADMISSION ONWARDS)

Duration: 1 Hour **Maximum Marks:** 35

Part A – Multiple Choice Questions

Answer all questions

Each question carries 1 mark

1. Where does most nutrient absorption take place?
   ANS: c) Small Intestine [K]/[CO1]

2. Which company is known for producing genetically modified crops?
   ANS c) Monsanto [K]/[CO5]
3. Which of the following belongs to traditional Nutraceuticals?
   ANS d) All of these [U]/[CO2]
4. Justify what does marketing primarily focus on
   ANS : b) Strategy to promote a brand or product [E]/[CO4]
5. Which category of nutraceuticals includes capsules, tablets, powders, and liquids with essential nutrients?
   ANS : c) Dietary supplements [K]/[CO2]

Part B – Fill in the Blanks

Answer all questions

Each question carries 2 marks

6. _____ is a spice that some people use to help lower cholesterol whereas _____ is a plant used to soothe sunburns. [K]/[CO1]
ANS: GarlicTurmeric,Cinnamon, Fenugreek, Coriander, Cinnamom (any one) Aloe Vera

7. _____ is a phytochemical found in marine sources like starfish. [K]/[CO2]
ANS: For attempt give full marks

8. Name two key factors driving the growth of the nutraceutical industry. [E]/[CO5]
ANS: : Increasing health awareness , Rising prevalence of lifestyle diseases (Any words Related to these)

9. _____ is a herbal product that combines honey with ashwagandha to support immune and digestive health. [U]/[CO5]
ANS: Dabur Ashwagandha Honey

10. Cleanroom environments are used to minimize the risk of _____. [U]/[CO3]
ANS: Contamination

Part C – Short Answer Questions

Answer any 5 questions

Each question carries 3 marks

11. Recall the causes and symptoms of respiratory disorders. [K]/[CO1]
ANS: Causes of respiratory disorders – 1.5marks
Symptoms of respiratory disorders – 1.5marks

12. Commonly found minerals in multivitamins. [An]/[CO2]
ANS: Mention any three minerals (Calcium, Iron , Zinc, potassium, Iron , Iodine, Selenium. Magnesium Manganese etc)

13. Explain the market expansion in nutraceuticals. [U]/[CO5]
ANS: Global Market Growth
India's Rapid Expansion
China's Contribution (one mark each)

14. What are the major challenges in scaling up a nutraceutical prototype for mass production? [An]/[CO3]
ANS: Any 3 challenges (one mark each)

15. What is the name of the tea that combines organic tulsi and green tea? [U]/[CO5]
ANS: Organic India Tulsi Green Tea

16. Define social entrepreneur. Name different types. [U]/[CO5]
ANS: Definition 1 mark
1. The community social entrepreneurs
2. The nonprofit social entrepreneurs
3. The transformational social entrepreneurs
4. The Global social entrepreneurs (2marks)

17. How does calcium help the human body? Name one dietary source. [U]/[CO2]
ANS: Functions 2 marks
Source -1mark (Milk/diary products . Leafy vegetables etc)

Part D – Short Essay Questions

Answer any 1 question

Each question carries 5 marks

18. How do fortified and recombinant nutraceuticals differ? Provide examples. [K]/[CO1]
ANS: Any 4 Difference (4 marks)
Examples -1 mark

19. Describe FDA approval process in detail. Explain labelling provisions for nutraceuticals under FSSAI. [U]/[CO4

ANS: Explain FDA approvals

Discovery and development..

Preclinical research.

Clinical research.

FDA review. (3marks)

Labelling provisions for FSSAI:

(i) The labelling on the nutraceutical-based food items shall not claim that the packed food items offer medicinal benefits or treat any chronic disease.

(ii) The statement by the manufacturer related to the function or structure or the general well-being of the body may be permitted by the FSSAI if the legit scientific data support the statement;

(iii) Every package of nutraceutical-based food shall enclose the given details on the label, namely: the word "NUTRACEUTICAL"; the common name of the nutraceutical;

(2marks)